

Kei Imada

Pronounced like the letter after “J”

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ABSTRACT

Third year Ph. D. student at **Cornell University**, developing discrete verification tools that prove temporal, probabilistic properties of continuous-time systems. Former **networking software engineer**.

Skills: Python, Golang, C, C++, OCaml, $\text{\LaTeX}$, Z3, TLA+, MPI, CUDA, React, Typescript, SQL, Japanese, and Chinese.

PUBLICATIONS

A Nonstandard Temporal Logic for Continuous-Time Verification

In Submission

Kei Imada, Kazuki Watanabe, Justin Hsu, and Kohei Suenaga

ACADEMICS

Cornell University in Ithaca, NY

August 2023 → Present

Doctor of Philosophy in Computer Science

Advisor: Justin Hsu

• Kleene Algebra, Category Theory, Compilers, Program Synthesis, Formal Methods in Robotics

4.11 GPA

Swarthmore College in Swarthmore, PA

August 2016 → May 2020

Bachelor of Arts with dual majors in Computer Science and Mathematics

Cumulative 3.94 and Major 3.95 GPA

Thesis: The Equivalence of Typed λ Calculi and Cartesian Closed Categories

Phi Beta Kappa and Sigma Xi

• Parallel and Distributed Computing, Programming Languages, Algorithms, Networks, Artificial Intelligence, Natural Language Processing, Category Theory, Topology, Real Analysis, Modern Algebra, Differential Equations, Modeling

Budapest Semesters in Mathematics in Budapest, Hungary

January 2019 → May 2019

Magas Kitüntetéssel High Honors (A or above on 5+ courses)

4.0 GPA

• Real Functions and Measures, Theory of Computing, Conjecture & Proof, Topology, Mathematical Cryptography

EXPERIENCE

Teaching Assistant at *Cornell University* in Ithaca, NY

August 2025 → Present

- Instructed 30 students in *CS 6120: Advanced Compilers* on common compiler optimizations
- Monitored and coached students through Zulip, an online discussion platform
- Graded biweekly assignments and provided feedback on code and design choices

Research Assistant at *Cornell University* in Ithaca, NY

August 2023 → Present

- Developed a logic to prove temporal properties about hybrid systems using nonstandard analysis
- Built a programming language using *menhir* to model hybrid systems and prove loop invariants with Z3
- Proved critical safety properties of simplified train control and water-level monitor systems

Research Intern at *National Institute of Informatics* in Tokyo, Japan

February 2025 → August 2025

- Collaborated with Ichiro Hasuo and Kazuki Watanabe on developing a program logic for continuous-time verification
- Presented preliminary results at CSCAT 2025 and Kyoto University

Software Engineer at *Pure Storage* in Mountain View, CA

August 2020 → August 2023

- Developed an internal firewall library with in-memory caching and batch-commit functionalities for performance
- Improved an existing Python *iptables* library's performance by 2 orders of magnitude
- Integrated *iptables* into FlashBlade clusters and improved its initialization and sync performance
- Implemented VLAN translation with field processor rules into Broadcom Tomahawk3 chips
- Spearheaded a formal verification project to secure FlashBlade's networking stacks
- Triaged over 500 test failures and discovered zero-day vulnerabilities and race conditions
- Organized and presented a 60-minute tech talk on testbed architectures and connectivity failures for over 80 engineers

Software Engineer Intern at *Pure Storage* in Mountain View, CA

June 2019 → August 2019

- Designed and implemented a scalable web analytics tool that detects and diagnoses SSD drive failures
- Improved latency by 2-fold with Redis caching layers storing structured responses from Amazon Redshift
- Developed the frontend using React and Typescript, with *ag-grid*, *highcharts*, and *react-select* as core components

Project Lead at *Swarthmore College Computer Society (SCCS)*

January 2018 → May 2020

Airpool at *SCCS*

September 2018

- Developed a website that streamlined carpooling between Swarthmore and popular transportation hubs
- Scheduled more than 200 rides which saved over \$2,000 in transportation costs

TriCo Course Scheduler at *SCCS*

May 2017

- Developed a website that assisted over 1,000 students schedule their courses out of over 10,000 courses

- Awarded a \$5,000 scholarship at a Swarthmore College hackathon as the Best Educational Hack

SINGING

Repertoire: My Way, Fly Me To The Moon, My Funny Valentine, Unchained Melody, 千里之外, and more

- Awarded a semesterly \$560 scholarship to take classical singing lessons with Professor Nancy Jantsch
- A high baritone at Swarthmore College Choir for 7 semesters